

Volume I Set Theory Reference

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Set Theory

0.1 Model of Set Theory

0.2 Sets

0.2.1 Sets, Membership, and ZFC Axioms

Definition 0.2.1 (Set and Membership). In axiomatic set theory, the notions of *set* and *membership* are *primitive*.

- A *set* is an object.
- *Membership* is a binary relation, denoted by $\in$, between objects.

If x is an object and A is a set, the statement $x \in A$ is read as “ x is an element of A .” No definition of “set” or “ $\in$ ” is given in more basic terms. Their meaning is determined entirely by the axioms governing them.

0.2.2 Set Constructions and Operations

Definition 0.2.2 (Empty Set). The unique set with no elements is called the *empty set* and is denoted $\emptyset$.

Definition 0.2.3 (Subset). Let A and B be sets. We say A is a *subset* of B , written $A \subseteq B$, if every element of A is also an element of B :

$$A \subseteq B \iff \forall x (x \in A \rightarrow x \in B).$$

Definition 0.2.4 (Proper Subset). A is a *proper subset* of B , written $A \subsetneq B$, if $A \subseteq B$ and $A \neq B$.

Definition 0.2.5 (Set Equality). Two sets A and B are *equal*, written $A = B$, if they have the same elements:

$$A = B \iff \forall x (x \in A \leftrightarrow x \in B).$$

Equivalently, $A = B \iff (A \subseteq B \wedge B \subseteq A)$.

Definition 0.2.6 (Union). The *union* of A and B , denoted $A \cup B$, is the set of all elements belonging to at least one of the two sets:

$$A \cup B = \{x \mid x \in A \vee x \in B\}.$$

Definition 0.2.7 (Intersection). The *intersection* of A and B , denoted $A \cap B$, is the set of all elements common to both:

$$A \cap B = \{x \mid x \in A \wedge x \in B\}.$$

Definition 0.2.8 (Set Difference). The *set difference* $A \setminus B$ is the set of elements in A but not in B :

$$A \setminus B = \{x \mid x \in A \wedge x \notin B\}.$$

Definition 0.2.9 (Symmetric Difference). The *symmetric difference* $A \triangle B$ is the set of elements belonging to exactly one of A and B :

$$A \triangle B = (A \setminus B) \cup (B \setminus A) = \{x \mid (x \in A \vee x \in B) \wedge x \notin A \cap B\}.$$

Definition 0.2.10 (Complement). Let U be a fixed universe and $A \subseteq U$. The *complement* of A relative to U , denoted A^c , is:

$$A^c = U \setminus A.$$

Definition 0.2.11 (Cartesian Product). The *Cartesian product* of sets A and B is:

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}.$$

Definition 0.2.12 (Power Set). The *power set* of A , denoted $\mathcal{P}(A)$, is the set of all subsets of A :

$$\mathcal{P}(A) := \{S \mid S \subseteq A\}.$$

Theorem 0.2.13 (De Morgan's Laws). Let U be a universe and $A, B \subseteq U$. Then

$$(A \cup B)^c = A^c \cap B^c \quad \text{and} \quad (A \cap B)^c = A^c \cup B^c.$$

Go to proof.

Definition 0.2.14 (Set Duality). Two set-theoretic expressions over a fixed universe U are *dual* if one is obtained from the other by simultaneously replacing $\cup \leftrightarrow \cap$ and $\emptyset \leftrightarrow U$, with complements unchanged.

Corollary 0.2.15 (Principle of Set Duality). Any identity involving $\cup$, $\cap$, $\emptyset$, and U that holds for all subsets of a universe remains valid when each operation and constant is replaced by its dual.

0.2.3 Algebraic Laws of Set Operations

Theorem 0.2.16 (Commutativity of Union and Intersection). Let A, B be sets. Then

$$A \cup B = B \cup A \quad \text{and} \quad A \cap B = B \cap A.$$

Go to proof.

Theorem 0.2.17 (Associativity of Union and Intersection). Let A, B, C be sets. Then

$$(A \cup B) \cup C = A \cup (B \cup C) \quad \text{and} \quad (A \cap B) \cap C = A \cap (B \cap C).$$

Go to proof.

Theorem 0.2.18 (Distributive Laws). Let A, B, C be sets. Then

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C),$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

Go to proof.

Theorem 0.2.19 (Identity and Absorption Laws). Let A, B be sets and U a universe with $A \subseteq U$. Then

$$A \cup \emptyset = A, \quad A \cap U = A,$$

$$A \cup (A \cap B) = A, \quad A \cap (A \cup B) = A.$$

Go to proof.

Theorem 0.2.20 (Involution of Complement). For any $A \subseteq U$,

$$(A^c)^c = A.$$

Go to proof.

0.3 Families

0.3.1 Indexed Families of Sets

Definition 0.3.1 (Indexed Family of Sets). Let I be a set (the *index set*) and U a universe. An *indexed family of sets* is a function

$$F : I \rightarrow \mathcal{P}(U),$$

with $F(i)$ typically written A_i . The family is denoted $\{A_i\}_{i \in I}$.

Definition 0.3.2 (Indexed Union). The *union* of the indexed family $\{A_i\}_{i \in I}$ is:

$$\bigcup_{i \in I} A_i := \{x \mid \exists i \in I \text{ such that } x \in A_i\}.$$

Definition 0.3.3 (Indexed Intersection). For $I \neq \emptyset$, the *intersection* of $\{A_i\}_{i \in I}$ is:

$$\bigcap_{i \in I} A_i := \{x \mid \forall i \in I, x \in A_i\}.$$

Theorem 0.3.4 (De Morgan's Laws for Indexed Families). Let $\{A_i\}_{i \in I}$ be an indexed family of subsets of a universe U . Then

$$U \setminus \bigcup_{i \in I} A_i = \bigcap_{i \in I} (U \setminus A_i).$$

If $I \neq \emptyset$, then

$$U \setminus \bigcap_{i \in I} A_i = \bigcup_{i \in I} (U \setminus A_i).$$

Go to proof.

Definition 0.3.5 (Pairwise Disjoint Family). The family $\{A_i\}_{i \in I}$ is *pairwise disjoint* if

$$\forall i, j \in I, i \neq j \rightarrow A_i \cap A_j = \emptyset.$$

Definition 0.3.6 (Cover). A collection $\mathcal{C} \subseteq \mathcal{P}(A)$ is a *cover* of A if

$$\bigcup_{C \in \mathcal{C}} C = A.$$

Arbitrary Cartesian Products

Definition 0.3.7 (Arbitrary Cartesian Product). Let $\{A_i\}_{i \in I}$ be an indexed family of sets. The *Cartesian product* is:

$$\prod_{i \in I} A_i := \left\{ f : I \rightarrow \bigcup_{i \in I} A_i \mid \forall i \in I, f(i) \in A_i \right\}.$$

0.3.2 Covers, Subcovers, and the Finite Intersection Property

Definition 0.3.8 (Cover). Let A be a set and $\mathcal{C}$ a collection of sets. $\mathcal{C}$ is a *cover* of A if every element of A belongs to at least one member of $\mathcal{C}$:

$$A \subseteq \bigcup_{C \in \mathcal{C}} C.$$

Definition 0.3.9 (Subcover). $\mathcal{C}'$ is a *subcover* of $\mathcal{C}$ if $\mathcal{C}' \subseteq \mathcal{C}$ and $\mathcal{C}'$ is still a cover of A .

Definition 0.3.10 (Finite Cover). A cover is *finite* if it has finitely many members. A *finite subcover* is a subcover $\mathcal{C}' \subseteq \mathcal{C}$ with $|\mathcal{C}'| < \infty$.

Definition 0.3.11 (Open Cover). In a topological space (X, τ) , an *open cover* of a subset $A \subseteq X$ is a cover $\mathcal{C}$ of A all of whose members are open sets: $\mathcal{C} \subseteq \tau$.

Definition 0.3.12 (Finite Intersection Property). A collection $\mathcal{F}$ of sets has the *finite intersection property* (FIP) if every finite subcollection has nonempty intersection: for all finite $\mathcal{F}' \subseteq \mathcal{F}$,

$$\bigcap_{F \in \mathcal{F}'} F \neq \emptyset.$$

Proposition 0.3.13 (FIP–Cover Duality). *Let X be a set, $A \subseteq X$, and $\{U_\alpha\}_{\alpha \in I}$ a family of subsets of X . Let $F_\alpha = U_\alpha^c = X \setminus U_\alpha$. Then:*

$$\{U_\alpha\} \text{ covers } A \iff \{F_\alpha \cap A\}_{\alpha \in I} \text{ does not have the FIP.}$$

More precisely: the family $\{U_\alpha\}$ does not cover A if and only if the family of closed complements $\{F_\alpha\}$ has the FIP relative to A . Go to proof.

Proofs

Relations

0.4 Relations — Quick Reference

0.4.1 Ordered Pairs and Relations

Definition 0.4.1 (Ordered Pair). Let a and b be sets. The *ordered pair* (a, b) is defined (Kuratowski) as

$$(a, b) := \{\{a\}, \{a, b\}\}.$$

Theorem 0.4.2 (Uniqueness of Ordered Pairs). *For any sets a, b, c, d ,*

$$(a, b) = (c, d) \iff (a = c \wedge b = d).$$

Go to proof.

Definition 0.4.3 (Cartesian Product — as foundation for relations). The *Cartesian product* of sets A and B is:

$$A \times B := \{(a, b) \mid a \in A \text{ and } b \in B\}.$$

Definition 0.4.4 (Relation). A *relation* from A to B is any subset $R \subseteq A \times B$.

If $(a, b) \in R$ we write $a R b$ and say “ a is related to b .” When $A = B$ we say R is a *relation on A* .

Definition 0.4.5 (Domain of a Relation). Let $R \subseteq A \times B$ be a relation. The *domain* of R is

$$\text{Dom}(R) := \{a \in A \mid \exists b \in B, (a, b) \in R\}.$$

Definition 0.4.6 (Range of a Relation). Let $R \subseteq A \times B$ be a relation. The *range* of R is

$$\text{Ran}(R) := \{b \in B \mid \exists a \in A, (a, b) \in R\}.$$

Tarski also calls this class the *counter-domain* or *converse domain*.

Definition 0.4.7 (Converse Relation). If $R \subseteq A \times B$, the *converse* of R is the relation $R^{-1} \subseteq B \times A$ defined by

$$(b, a) \in R^{-1} \iff (a, b) \in R.$$

Equivalently, $b R^{-1} a$ iff $a R b$.

Definition 0.4.8 (Identity Relation). For a set A , the *identity relation* on A is

$$\Delta_A := \{(a, a) \mid a \in A\}.$$

Definition 0.4.9 (Composition of Relations). Let $R \subseteq A \times B$ and $S \subseteq B \times C$. The *composition* $S \circ R \subseteq A \times C$ is defined by

$$(a, c) \in S \circ R \iff \exists b \in B, (a, b) \in R \text{ and } (b, c) \in S.$$

Theorem 0.4.10 (Associativity of Relation Composition). *Let $R \subseteq A \times B$, $S \subseteq B \times C$, and $T \subseteq C \times D$. Then*

$$T \circ (S \circ R) = (T \circ S) \circ R.$$

Go to proof.

Theorem 0.4.11 (Identity Laws for Relation Composition). *Let $R \subseteq A \times B$. Then*

$$R \circ \Delta_A = R \quad \text{and} \quad \Delta_B \circ R = R.$$

Go to proof.

Definition 0.4.12 (Null Relation). For sets A and B , the *null relation* from A to B is $\emptyset \subseteq A \times B$. No element of A is related to any element of B .

Definition 0.4.13 (Universal Relation). For sets A and B , the *universal relation* from A to B is $A \times B$ itself. Every element of A is related to every element of B .

Definition 0.4.14 (Relation Inclusion). Let $R, S \subseteq A \times B$. Relation inclusion is ordinary set inclusion:

$$R \subseteq S \iff \forall a \in A \forall b \in B, ((a, b) \in R \Rightarrow (a, b) \in S),$$

Definition 0.4.15 (Relation Equality). Let $R, S \subseteq A \times B$. Relation equality is ordinary set equality:

$$R = S \iff R \subseteq S \text{ and } S \subseteq R.$$

Definition 0.4.16 (Union of Relations). Let $R, S \subseteq A \times B$. The *union* $R \cup S$ is the relation with

$$(a, b) \in R \cup S \iff (a, b) \in R \text{ or } (a, b) \in S.$$

Definition 0.4.17 (Intersection of Relations). Let $R, S \subseteq A \times B$. The *intersection* $R \cap S$ is the relation with

$$(a, b) \in R \cap S \iff (a, b) \in R \text{ and } (a, b) \in S.$$

Definition 0.4.18 (Difference of Relations). Let $R, S \subseteq A \times B$. The *difference* $R \setminus S$ is the relation with

$$(a, b) \in R \setminus S \iff (a, b) \in R \text{ and } (a, b) \notin S.$$

Definition 0.4.19 (Complement of a Relation). Let $R \subseteq A \times B$. The *complement* of R relative to $A \times B$ is

$$(A \times B) \setminus R.$$

Theorem 0.4.20 (Converse Laws for Relation Operations). *Let $R, S \subseteq A \times B$, let $T \subseteq B \times C$, and take complements relative to the indicated Cartesian products. Then*

$$\begin{aligned} (R^{-1})^{-1} &= R, & (T \circ R)^{-1} &= R^{-1} \circ T^{-1}, \\ (R \cup S)^{-1} &= R^{-1} \cup S^{-1}, & (R \cap S)^{-1} &= R^{-1} \cap S^{-1}, \\ (R \setminus S)^{-1} &= R^{-1} \setminus S^{-1}, & ((A \times B) \setminus R)^{-1} &= (B \times A) \setminus R^{-1}. \end{aligned}$$

Go to proof.

Theorem 0.4.21 (Relation Composition and Boolean Operations). *Let $P, Q \subseteq A \times B$, $R \subseteq B \times C$, and $S, T \subseteq B \times C$. Then*

$$R \circ (P \cup Q) = (R \circ P) \cup (R \circ Q), \quad (S \cup T) \circ P = (S \circ P) \cup (T \circ P).$$

Also,

$$R \circ (P \cap Q) \subseteq (R \circ P) \cap (R \circ Q), \quad (S \cap T) \circ P \subseteq (S \circ P) \cap (T \circ P).$$

The analogous formulas for difference and complement hold only as inclusions or under extra hypotheses; they are not general distributive laws. Go to proof.

Definition 0.4.22 (Transitive Closure). Let R be a relation on A . The *transitive closure* of R is the smallest transitive relation on A containing R .

Definition 0.4.23 (Equivalence Closure). Let R be a relation on A . The *equivalence closure* of R is the smallest equivalence relation on A containing R .

Definition 0.4.24 (One-to-Many Relation). Let $R \subseteq A \times B$. R is *one-to-many* if some input is related to two distinct outputs:

$$\exists a \in A \exists b_1, b_2 \in B, \quad (a, b_1) \in R \wedge (a, b_2) \in R \wedge b_1 \neq b_2.$$

Definition 0.4.25 (Many-to-One Relation). Let $R \subseteq A \times B$. R is *many-to-one* if two distinct inputs are related to a common output:

$$\exists a_1, a_2 \in A \exists b \in B, \quad (a_1, b) \in R \wedge (a_2, b) \in R \wedge a_1 \neq a_2.$$

Definition 0.4.26 (Many-to-Many Relation). Let $R \subseteq A \times B$. R is *many-to-many* if it is one-to-many and many-to-one.

Definition 0.4.27 (Many-Place Relation). A *ternary relation* among sets A , B , and C is a subset of $A \times B \times C$. More generally, an n -place relation on $A_1, \dots, A_n$ is a subset of

$$A_1 \times \dots \times A_n.$$

For example, betweenness in geometry is naturally a three-place relation, and the arithmetic statement $x + y = z$ is naturally a three-place relation among x , y , and z .

0.4.2 Properties of Relations

Definition 0.4.28 (Reflexive Relation). A binary relation $R \subseteq A \times A$ is *reflexive* on A if every element of the domain is related to itself:

$$\forall a \in A, \quad (a, a) \in R.$$

In infix notation, this is the condition $\forall a \in A, \quad aRa$.

Definition 0.4.29 (Irreflexive Relation). A binary relation $R \subseteq A \times A$ is *irreflexive* on A if no element of the domain is related to itself:

$$\forall a \in A, \quad (a, a) \notin R.$$

In infix notation, this is the condition $\forall a \in A, \quad \neg(aRa)$.

Definition 0.4.30 (Symmetric Relation). A binary relation $R \subseteq A \times A$ is *symmetric* on A if every related pair also appears in the reverse order:

$$\forall a, b \in A, \quad (a, b) \in R \rightarrow (b, a) \in R.$$

In infix notation, this is $\forall a, b \in A, \quad aRb \rightarrow bRa$.

Definition 0.4.31 (Antisymmetric Relation). A binary relation $R \subseteq A \times A$ is *antisymmetric* on A if any two elements related in both directions must be equal:

$$\forall a, b \in A, \quad ((a, b) \in R \wedge (b, a) \in R) \rightarrow a = b.$$

In infix notation, this is $\forall a, b \in A, \quad ((aRb \wedge bRa) \rightarrow a = b)$.

Definition 0.4.32 (Asymmetric Relation). A binary relation $R \subseteq A \times A$ is *asymmetric* on A if every related pair excludes the reverse pair:

$$\forall a, b \in A, \quad (a, b) \in R \rightarrow (b, a) \notin R.$$

In infix notation, this is $\forall a, b \in A, \quad (aRb \rightarrow \neg(bRa))$.

Definition 0.4.33 (Transitive Relation). A binary relation $R \subseteq A \times A$ is *transitive* on A if any two composable related pairs collapse to a direct related pair:

$$\forall a, b, c \in A, \quad ((a, b) \in R \wedge (b, c) \in R) \rightarrow (a, c) \in R.$$

In infix notation, this is $\forall a, b, c \in A, \quad ((aRb \wedge bRc) \rightarrow aRc)$.

Definition 0.4.34 (Total (Connex) Relation). R is *total* (or *connex*) if every pair is comparable:

$$\forall a, b \in A, (a, b) \in R \vee (b, a) \in R.$$

Definition 0.4.35 (Equivalence Relation). R is an *equivalence relation* on A if it is reflexive, symmetric, and transitive.

Definition 0.4.36 (Preorder). R is a *preorder* on A if it is reflexive and transitive.

Definition 0.4.37 (Partial Order). R is a *partial order* on A if it is reflexive, antisymmetric, and transitive.

Definition 0.4.38 (Strict Partial Order). R is a *strict partial order* on A if it is irreflexive and transitive.

Definition 0.4.39 (Total Order). R is a *total order* on A if it is a partial order and is total.

0.5 Equivalence

0.5.1 Equivalence Classes and Partitions

Definition 0.5.1 (Equivalence Class). Let R be an equivalence relation on A . For $a \in A$, the *equivalence class* of a is:

$$[a]_R := \{x \in A \mid (a, x) \in R\}.$$

When R is clear from context, we write $[a]$.

Definition 0.5.2 (Quotient Set). The *quotient set* of A by R is the set of all equivalence classes:

$$A/R := \{[a]_R \mid a \in A\}.$$

Definition 0.5.3 (Index of an Equivalence Relation). The *index* of R on A is the cardinality $|A/R|$, i.e. the number of equivalence classes.

Definition 0.5.4 (Canonical Surjection). The *canonical surjection* (quotient map) is the function

$$\pi : A \rightarrow A/R, \quad \pi(a) := [a].$$

Definition 0.5.5 (Partition). A *partition* of A is a collection $\mathcal{P}$ of subsets of A such that:

- (i) every block is nonempty: $\forall P \in \mathcal{P}, P \neq \emptyset$;
- (ii) distinct blocks are disjoint: $\forall P, Q \in \mathcal{P}, P \neq Q \rightarrow P \cap Q = \emptyset$;
- (iii) the blocks cover A : $\bigcup_{P \in \mathcal{P}} P = A$.

The sets $P \in \mathcal{P}$ are called the *blocks* of the partition.

Lemma 0.5.6 (Representative Independence Lemma). *Let R be an equivalence relation on A . For any $a, b \in A$,*

$$[a] = [b] \iff (a, b) \in R.$$

Go to proof.

Theorem 0.5.7 (Equivalence Relations and Partitions). *Let A be a set.*

- (i) *If R is an equivalence relation on A , then A/R is a partition of A .*
- (ii) *If $\mathcal{P}$ is a partition of A , then the relation $R_{\mathcal{P}}$ defined by*

$$(a, b) \in R_{\mathcal{P}} \iff \exists P \in \mathcal{P} \text{ with } a \in P \text{ and } b \in P$$

is an equivalence relation on A .

(iii) These constructions are inverse: $R_{A/R} = R$ and $A/R_{\mathcal{P}} = \mathcal{P}$.

Go to proof.

Theorem 0.5.8 (Universal Property of the Quotient Map). *Let R be an equivalence relation on A , let $\pi : A \rightarrow A/R$ be the canonical surjection, and let $f : A \rightarrow B$ be any function. Then f is constant on equivalence classes — meaning $(a, b) \in R$ implies $f(a) = f(b)$ — if and only if there exists a unique function $\bar{f} : A/R \rightarrow B$ such that*

$$f = \bar{f} \circ \pi.$$

When it exists, $\bar{f}$ is defined by $\bar{f}([a]) = f(a)$. Go to proof.

Proofs

Functions

0.6 Functions — Quick Reference

0.6.1 Functions

Definition 0.6.1 (Functional Relation). Let $R \subseteq A \times B$ be a relation. We say that R is *functional from A to B* if each input has at most one output:

$$\forall a \in A \forall b_1, b_2 \in B, \quad ((a, b_1) \in R \wedge (a, b_2) \in R) \Rightarrow b_1 = b_2.$$

Definition 0.6.2 (Total Relation on a Domain). Let $R \subseteq A \times B$ be a relation. We say that R is *total on A* if each input has at least one output:

$$\forall a \in A \exists b \in B, \quad (a, b) \in R.$$

Definition 0.6.3 (Function). Let A and B be sets. A *function* from A to B is a relation $f \subseteq A \times B$, together with the declared domain A and codomain B , such that:

- (i) (*Existence / left-total*) for every $a \in A$, there exists $b \in B$ with $(a, b) \in f$;
- (ii) (*Uniqueness / right-unique*) if $(a, b_1) \in f$ and $(a, b_2) \in f$ then $b_1 = b_2$.

If $(a, b) \in f$ we write $f(a) = b$.

Theorem 0.6.4 (Functional Relation Characterization of Functions). *Let A and B be sets, and let $R \subseteq A \times B$. Then R is a function from A to B if and only if*

$$\forall a \in A \exists! b \in B, \quad (a, b) \in R.$$

Equivalently, R is a function from A to B if and only if R is total on A and functional from A to B . Go to proof.

Definition 0.6.5 (Partial Function). Let A and B be sets. A *partial function* from A to B is a functional relation $R \subseteq A \times B$. It need not be total on A : some inputs may have no output.

Proposition 0.6.6 (Partial Functions Are Functional Relations). *For sets A and B and a relation $R \subseteq A \times B$, R is a partial function from A to B if and only if R is functional from A to B . Go to proof.*

Definition 0.6.7 (Function Equality). For functions $f : A \rightarrow B$ and $g : C \rightarrow D$, we write $f = g$ if

$$A = C, \quad B = D, \quad \text{and} \quad \forall a \in A, f(a) = g(a).$$

Proposition 0.6.8 (One-to-Many Relations Are Not Functional). *If $R \subseteq A \times B$ is one-to-many, then R is not functional from A to B . Consequently, R is not a function from A to B . Go to proof.*

Definition 0.6.9 (Domain of a Function). If f is a function from A to B , we write $f : A \rightarrow B$, where A is the *domain* $\text{dom}(f)$.

Definition 0.6.10 (Codomain of a Function). If f is a function from A to B , we write $f : A \rightarrow B$, where B is the *codomain* $\text{cod}(f)$.

Definition 0.6.11 (Image of a Function). The *image* (or *range*) of $f : A \rightarrow B$ is:

$$\text{im}(f) := \{b \in B \mid \exists a \in A, f(a) = b\}.$$

The image may be a proper subset of the codomain.

Definition 0.6.12 (Image of a Set). For $S \subseteq A$, the *image of S under f* is:

$$f(S) := \{f(a) \mid a \in S\}.$$

Note $f(A) = \text{im}(f)$.

Definition 0.6.13 (Preimage). For $T \subseteq B$, the *preimage* (inverse image) of T under f is:

$$f^{-1}(T) := \{a \in A \mid f(a) \in T\}.$$

Definition 0.6.14 (Fiber). The *fiber* of f over $b \in B$ is the preimage of the singleton:

$$f^{-1}(\{b\}) = \{a \in A \mid f(a) = b\}.$$

Definition 0.6.15 (Graph of a Function). The *graph* of $f : A \rightarrow B$ is:

$$\text{Graph}(f) := \{(a, b) \in A \times B \mid b = f(a)\}.$$

Definition 0.6.16 (Injective Function). $f : A \rightarrow B$ is *injective* (one-to-one) if distinct elements of A have distinct images:

$$\forall a_1, a_2 \in A, f(a_1) = f(a_2) \implies a_1 = a_2.$$

Definition 0.6.17 (Empty Function). For every set B , the *empty function* from $\emptyset$ to B is the empty relation

$$\emptyset : \emptyset \rightarrow B.$$

It is the unique function with domain $\emptyset$ and codomain B .

Definition 0.6.18 (Surjective Function). $f : A \rightarrow B$ is *surjective* (onto) if every element of B is achieved:

$$\forall b \in B, \exists a \in A \text{ such that } f(a) = b.$$

Equivalently, $\text{im}(f) = B$.

Definition 0.6.19 (Bijective Function). $f : A \rightarrow B$ is *bijective* if it is both injective and surjective.

Proposition 0.6.20 (One-One Relations and Converse Functionality). Let $f : A \rightarrow B$ be a function, identified with its graph $G_f \subseteq A \times B$. Then f is injective if and only if the converse relation $G_f^{-1} \subseteq B \times A$ is functional from B to A . Go to proof.

Proposition 0.6.21 (Bijections Have Unique Correspondence Both Ways). Let $f : A \rightarrow B$ be a function. Then f is bijective if and only if

$$\forall a \in A \exists! b \in B, f(a) = b,$$

and

$$\forall b \in B \exists! a \in A, f(a) = b.$$

Go to proof.

Definition 0.6.22 (Many-Place Functional Relation). A relation $R \subseteq A_1 \times \cdots \times A_n \times B$ is *functional in the last coordinate* if

$$\begin{aligned} &\forall a_1 \in A_1 \cdots \forall a_n \in A_n \forall b_1, b_2 \in B, \\ &((a_1, \dots, a_n, b_1) \in R \wedge (a_1, \dots, a_n, b_2) \in R) \implies b_1 = b_2. \end{aligned}$$

When such a relation is also total in the first n coordinates, it determines a function of n variables, written $f(a_1, \dots, a_n) = b$.

Definition 0.6.23 (Identity Function). The *identity function* on A is $\text{id}_A : A \rightarrow A$, $\text{id}_A(a) = a$.

Definition 0.6.24 (Inclusion Map). If $A \subseteq B$, the *inclusion map* is $\iota : A \hookrightarrow B$, $\iota(a) = a$.

Definition 0.6.25 (Constant Function). $f : A \rightarrow B$ is *constant* if there exists $b_0 \in B$ with $f(a) = b_0$ for all $a \in A$.

0.6.2 Composition, Inverses, and Set-Image Laws

Definition 0.6.26 (Composition). Let $f : A \rightarrow B$ and $g : B \rightarrow C$. The *composition* $g \circ f : A \rightarrow C$ is:

$$(g \circ f)(a) := g(f(a)) \quad \text{for all } a \in A.$$

Theorem 0.6.27 (Associativity of Composition). For $f : A \rightarrow B$, $g : B \rightarrow C$, $h : C \rightarrow D$:

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

Go to proof.

Theorem 0.6.28 (Identity and Composition). For any $f : A \rightarrow B$:

$$f \circ \text{id}_A = f \quad \text{and} \quad \text{id}_B \circ f = f.$$

Go to proof.

Theorem 0.6.29 (Injectivity and Surjectivity Under Composition). Let $f : A \rightarrow B$ and $g : B \rightarrow C$.

- (i) If f and g are injective, then $g \circ f$ is injective.
- (ii) If f and g are surjective, then $g \circ f$ is surjective.
- (iii) If $g \circ f$ is injective, then f is injective.
- (iv) If $g \circ f$ is surjective, then g is surjective.

Go to proof.

Definition 0.6.30 (Inverse Function). Let $f : A \rightarrow B$ be bijective. The *inverse function* is $f^{-1} : B \rightarrow A$ defined by: $f^{-1}(b) = a$ iff $f(a) = b$.

Theorem 0.6.31 (Characterization of Inverse Functions). Let $f : A \rightarrow B$ be bijective. Then

$$f^{-1} \circ f = \text{id}_A \quad \text{and} \quad f \circ f^{-1} = \text{id}_B.$$

Conversely, a function admits an inverse iff it is bijective. Go to proof.

Theorem 0.6.32 (Inverse of a Composition). Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be bijective. Then $g \circ f$ is bijective and

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}.$$

Go to proof.

Definition 0.6.33 (Left Inverse). Let $f : A \rightarrow B$. A *left inverse* of f is a function $g : B \rightarrow A$ with $g \circ f = \text{id}_A$.

Definition 0.6.34 (Right Inverse). Let $f : A \rightarrow B$. A *right inverse* (or *section*) of f is a function $h : B \rightarrow A$ with $f \circ h = \text{id}_B$.

Theorem 0.6.35 (One-Sided Inverses and Function Properties). Let $f : A \rightarrow B$.

- (i) f has a left inverse $\iff f$ is injective (and $A \neq \emptyset$ or $A = B = \emptyset$).
- (ii) f has a right inverse $\iff f$ is surjective.
- (iii) If f has both a left inverse g and a right inverse h , then $g = h$ and f is bijective.

Go to proof.

Definition 0.6.36 (Restriction). Let $f : A \rightarrow B$ and $C \subseteq A$. The *restriction* of f to C is $f|_C : C \rightarrow B$, $f|_C(a) = f(a)$ for all $a \in C$.

Definition 0.6.37 (Extension). Let $A \subseteq A'$, and let $f : A \rightarrow B$. A function $g : A' \rightarrow B$ is an *extension* of f if $g|_A = f$.

Theorem 0.6.38 (Preimages Preserve Set Operations). Let $f : A \rightarrow B$ and $S, T \subseteq B$. Then

- (i) $f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$,
- (ii) $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$,
- (iii) $f^{-1}(S \setminus T) = f^{-1}(S) \setminus f^{-1}(T)$,
- (iv) $f^{-1}(S^c) = (f^{-1}(S))^c$.

Go to proof.

Theorem 0.6.39 (Images and Set Operations). Let $f : A \rightarrow B$ and $S, T \subseteq A$. Then

- (i) $f(S \cup T) = f(S) \cup f(T)$,
- (ii) $f(S \cap T) \subseteq f(S) \cap f(T)$,
- (iii) $f(S \setminus T) \supseteq f(S) \setminus f(T)$.

Equality holds in (ii) and (iii) for all S, T iff f is injective. Go to proof.

Theorem 0.6.40 (Image–Preimage Adjunction). Let $f : A \rightarrow B$, let $S \subseteq A$, and let $T \subseteq B$. Then

$$S \subseteq f^{-1}(f(S)), \quad f(f^{-1}(T)) \subseteq T.$$

Moreover, $S = f^{-1}(f(S))$ for every $S \subseteq A$ iff f is injective, and $f(f^{-1}(T)) = T$ for every $T \subseteq B$ iff f is surjective. Go to proof.

Proofs

Orderings

0.7 Order

0.7.1 Ordered Sets

Definition 0.7.1 (Ordered Set). An *ordered set* is a set equipped with an order relation. Unless explicitly qualified, “ordered set” means a [partially ordered set](#), and its relation is written $\leq$.

Definition 0.7.2 (Partially Ordered Set). A *partially ordered set* is a pair $(A, \leq)$ where A is a set and $\leq$ is a [partial order](#) on A (a relation that is reflexive, antisymmetric, and transitive).

Definition 0.7.3 (Order-Structure Synonyms). The following short forms abbreviate their full names and carry no independent mathematical content; each resolves to the canonical structure it names:

- $poset :=$ [partially ordered set](#).
- $toset, loiset, linearly\ ordered\ set :=$ [totally ordered set](#) (total order and linear order coincide, so these name one structure).
- $woset :=$ [well-ordered set](#).

Definition 0.7.4 (Strict Order). Let $(A, \leq)$ be an ordered set. The *strict order* $<$ is defined by:

$$a < b \iff (a \leq b \wedge a \neq b).$$

Definition 0.7.5 (Comparable and Incomparable Elements). In an ordered set $(A, \leq)$, elements $a, b \in A$ are *comparable* if $a \leq b$ or $b \leq a$; they are *incomparable* if neither holds.

Definition 0.7.6 (Totally Ordered Set). A *totally ordered set* (or *linearly ordered set*) is a pair $(A, \leq)$ where $\leq$ is a [total order](#) on A — equivalently, a partially ordered set in which every pair of elements is [comparable](#):

$$\forall a, b \in A, a \leq b \vee b \leq a.$$

Definition 0.7.7 (Upper and Lower Bounds). Let $(A, \leq)$ be an ordered set and $S \subseteq A$.

An element $u \in A$ is an *upper bound* of S if $\forall s \in S, s \leq u$.

An element $\ell \in A$ is a *lower bound* of S if $\forall s \in S, \ell \leq s$.

Definition 0.7.8 (Minimal Element). Let $S \subseteq A$ in an ordered set $(A, \leq)$. An element $m \in S$ is a *minimal element* of S if there is no $s \in S$ with $s < m$.

Definition 0.7.9 (Maximal Element). Let $S \subseteq A$ in an ordered set $(A, \leq)$. An element $M \in S$ is a *maximal element* of S if there is no $s \in S$ with $M < s$.

Definition 0.7.10 (Least Element). Let $S \subseteq A$ in an ordered set $(A, \leq)$. An element $\ell \in S$ is the *least element* of S if

$$\forall s \in S, \ell \leq s.$$

Definition 0.7.11 (Greatest Element). Let $S \subseteq A$ in an ordered set $(A, \leq)$. An element $g \in S$ is the *greatest element* of S if

$$\forall s \in S, s \leq g.$$

Definition 0.7.12 (Order-Preserving Map). Let $(M, \leq)$ and $(M', \leq')$ be partially ordered sets. A function $f : M \rightarrow M'$ is *order-preserving* (or *monotone*) if

$$\forall a, b \in M, a \leq b \implies f(a) \leq' f(b).$$

Definition 0.7.13 (Order Isomorphism). Let $(M, \leq)$ and $(M', \leq')$ be partially ordered sets. A function $f : M \rightarrow M'$ is an *order isomorphism* if it is bijective and

$$\forall a, b \in M, a \leq b \iff f(a) \leq' f(b).$$

Definition 0.7.14 (Well-Ordered Set). A **totally ordered set** $(A, \leq)$ is *well-ordered* if every nonempty subset $S \subseteq A$ has a least element:

$$\forall S \subseteq A, (S \neq \emptyset \implies \exists m \in S \text{ s.t. } m \leq s \forall s \in S).$$

Definition 0.7.15 (Chain and Antichain). A subset $C \subseteq A$ of a poset $(A, \leq)$ is a *chain* if every pair of elements in C is comparable. It is an *antichain* if no two distinct elements of C are comparable.

Definition 0.7.16 (Initial Segment). A subset $I \subseteq A$ is an *initial segment* of $(A, \leq)$ if

$$a \in I \text{ and } b \leq a \implies b \in I.$$

Definition 0.7.17 (Directed Set). A preorder $(I, \leq)$ is *directed* (or a *directed set*) if every finite subset of I has an upper bound in I : for all $i, j \in I$ there exists $k \in I$ with $i \leq k$ and $j \leq k$.

Definition 0.7.18 (Net). Let $(I, \leq)$ be a directed set and X a set. A *net* in X is a function $x : I \rightarrow X$. The value at $i \in I$ is written x_i , and the net is written $(x_i)_{i \in I}$.

0.7.2 Supremum and Infimum

Definition 0.7.19 (Supremum and Infimum). Let $(A, \leq)$ be a poset and $S \subseteq A$.

The *supremum* (or *least upper bound*) of S , written $\sup S$, is an element $u^* \in A$ satisfying:

- (i) u^* is an upper bound of S : $\forall s \in S, s \leq u^*$;
- (ii) u^* is the *least* upper bound: if u is *any* upper bound of S , then $u^* \leq u$.

The *infimum* (or *greatest lower bound*) of S , written $\inf S$, is an element $\ell^* \in A$ satisfying:

- (i) ℓ^* is a lower bound of S : $\forall s \in S, \ell^* \leq s$;
- (ii) ℓ^* is the *greatest* lower bound: if ℓ is *any* lower bound of S , then $\ell \leq \ell^*$.

Proposition 0.7.20 (Uniqueness of Supremum and Infimum). Let $(A, \leq)$ be a poset and $S \subseteq A$. If $\sup S$ exists, it is unique. If $\inf S$ exists, it is unique.

Proposition 0.7.21 (Characterisation of Supremum). Let $(A, \leq)$ be a poset, $S \subseteq A$, and $u^* \in A$. Then $u^* = \sup S$ if and only if:

- (i) u^* is an upper bound of S ; and
- (ii) for every $v \in A$ with $v < u^*$, there exists $s \in S$ with $v < s$.

Proposition 0.7.22 (Duality of Supremum and Infimum). Let $(A, \leq)$ be a poset and $S \subseteq A$. Then $\inf_{(A, \leq)} S = \sup_{(A, \geq)} S$, where $(A, \geq)$ denotes the dual poset.

In particular: if every nonempty bounded-above subset of $(A, \leq)$ has a supremum, then every nonempty bounded-below subset of $(A, \leq)$ has an infimum.

0.7.3 Lattices

Definition 0.7.23 (Lattice). A *lattice* is a poset $(L, \leq)$ in which every two-element subset $\{a, b\} \subseteq L$ has both a supremum and an infimum in L .

The supremum $\sup\{a, b\}$ is called the *join* of a and b , written $a \vee b$. The infimum $\inf\{a, b\}$ is called the *meet* of a and b , written $a \wedge b$.

Definition 0.7.24 (Complete Lattice). A lattice $(L, \leq)$ is a *complete lattice* if every subset $S \subseteq L$ (including the empty set and L itself) has both a supremum and an infimum in L :

$$\forall S \subseteq L, \quad \sup S \in L \quad \text{and} \quad \inf S \in L.$$

Definition 0.7.25 (Distributive Lattice). A lattice $(L, \leq)$ is *distributive* if for all $a, b, c \in L$:

$$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c).$$

Definition 0.7.26 (Complement and Boolean Lattice). Let $(L, \leq)$ be a bounded distributive lattice with top element $\top$ and bottom element $\perp$. An element $a' \in L$ is a *complement* of $a \in L$ if

$$a \vee a' = \top \quad \text{and} \quad a \wedge a' = \perp.$$

A *Boolean lattice* (or *Boolean algebra*) is a bounded distributive lattice in which every element has a complement.

0.7.4 Hasse Diagrams and the Duality Principle

Definition 0.7.27 (Cover Relation). Let $(A, \leq)$ be a poset and $a, b \in A$. We say b *covers* a , written $a \lessdot b$, if $a < b$ and there is no $c \in A$ with $a < c < b$.

Definition 0.7.28 (Hasse Diagram). The *Hasse diagram* of a finite poset $(A, \leq)$ is the directed graph whose vertices are the elements of A , with an upward edge from a to b whenever $a \lessdot b$. Three graphical conventions apply:

- (i) Reflexive loops ($a \leq a$) are suppressed.
- (ii) Edges implied by transitivity are suppressed: if $a \lessdot b$ and $b \lessdot c$, no direct edge from a to c is drawn.
- (iii) Direction is encoded by height: $a \lessdot b$ is drawn with b strictly above a , so arrowheads are unnecessary.

Definition 0.7.29 (Dual Poset). Let $(A, \leq)$ be a poset. The *dual poset* is $(A, \geq)$, where $a \geq b$ is defined to mean $b \leq a$. The dual is also written $(A, \leq^{\text{op}})$.

Proposition 0.7.30 (Duality Principle for Posets). Let Φ be any first-order statement about a poset $(A, \leq)$ expressed using only the relation $\leq$. Let Φ^* be the statement obtained from Φ by replacing every occurrence of $\leq$ with $\geq$ (equivalently, working in the dual poset). Then:

$$(A, \leq) \models \Phi \iff (A, \geq) \models \Phi.$$

In particular, if Φ is a theorem about all posets, then so is Φ^* .

0.7.5 Order Extensions

Definition 0.7.31 (Symmetric and Asymmetric Parts). For any binary relation R on X , define:

- the *symmetric part*: $xIy \iff xRy \text{ and } yRx$
- the *asymmetric part*: $xPy \iff xRy \text{ but not } yRx$

Note that $R = P \cup I$.

Definition 0.7.32 (Maximal Elements and Upper Bounds). Let $\succsim$ be a binary relation on X .

- The set of *maximal elements* of X is

$$\text{Max}(X, \succsim) = \{x \in X \mid y \succsim x \text{ for no } y \in X \text{ with } y \neq x\}.$$

- The set of *upper bounds* of X is

$$M(X, \succsim) = \{x \in X \mid x \succsim y \text{ for all } y \in X\}.$$

Definition 0.7.33 (Extension of a Preorder). Let $\succsim$ be a preorder on X . A preorder $\succsim'$ is an *extension* of $\succsim$ if

$$x \succsim y \Rightarrow x \succsim' y, \quad x \succ y \Rightarrow x \succ' y.$$

A *complete extension* is an extension that is also complete.

Corollary 0.7.34 (Complete preorder extension). *For any nonempty set X and preorder $\succsim$ on X , there exists a complete preorder that is an extension of $\succsim$.*

Definition 0.7.35 (and Proposition (Only Weak Cycles)). Let $\succsim$ be a binary relation on X with asymmetric part $\succ$ and transitive closure $T(\succsim)$. We say $\succsim$ satisfies *only weak cycles* (OWC) if

$$xT(\succsim)y \Rightarrow \neg(y \succ x).$$

Proposition. A binary relation $\succsim$ on a nonempty set X can be extended to a complete preorder if and only if it satisfies OWC.

0.7.6 Induced Orders and Order Embeddings

Definition 0.7.36 (Induced Order). Let $(B, \leq')$ be a partially ordered set and let $f : A \rightarrow B$ be a function. The *order induced by f* (or *pullback order along f*) is the relation $\leq_f$ on A defined by:

$$x \leq_f y \iff f(x) \leq' f(y).$$

Proposition 0.7.37 ($\leq_f$ is always a preorder). *For any function $f : A \rightarrow B$ and partial order $(B, \leq')$, the relation $\leq_f$ is a preorder on A : it is reflexive and transitive.*

Definition 0.7.38 (f -Indistinguishable Elements). Let $f : A \rightarrow B$ and $\leq_f$ be the induced order. Two elements $x, y \in A$ are *f -indistinguishable*, written $x \sim_f y$, if both $x \leq_f y$ and $y \leq_f x$, i.e. if

$$f(x) \leq' f(y) \quad \text{and} \quad f(y) \leq' f(x).$$

Proposition 0.7.39 ($\leq_f$ is a partial order iff f is injective). *Let $(B, \leq')$ be a partially ordered set and $f : A \rightarrow B$. The induced order $\leq_f$ is a partial order on A if and only if f is injective.*

Definition 0.7.40 (Suborder / Restriction). Let $(B, \leq')$ be a partially ordered set and $S \subseteq B$. The *suborder* on S (or *induced order on S*) is the relation $\leq'_S$ defined by:

$$x \leq'_S y \iff x \leq' y, \quad \text{for } x, y \in S.$$

Definition 0.7.41 (Order Embedding). Let $(A, \leq)$ and $(B, \leq')$ be partially ordered sets. A function $f : A \rightarrow B$ is an *order embedding* if for all $x, y \in A$:

$$x \leq y \iff f(x) \leq' f(y).$$

Proposition 0.7.42 (Order embeddings are injective). *Every order embedding $f : (A, \leq) \rightarrow (B, \leq')$ is injective. Go to proof.*

Proposition 0.7.43 (Order embedding is isomorphism onto image). *If $f : (A, \leq) \rightarrow (B, \leq')$ is an order embedding, then f is an order isomorphism from $(A, \leq)$ to the suborder $(f(A), \leq'_{f(A)})$. Go to proof.*

0.8 Zorn

0.8.1 Zorn's Lemma and the Axiom of Choice

Proofs

Cardinality

0.9 Cardinality

0.9.1 Cardinality and Infinite Sets

Definition 0.9.1 (Same Cardinality). Two sets A and B have the *same cardinality*, written $A \sim B$, if there exists a bijection $f : A \rightarrow B$.

Definition 0.9.2 (Finite and Infinite Sets). A set A is *finite* if $A = \emptyset$ or $A \sim \{1, 2, \dots, n\}$ for some $n \in \mathbb{N}$. A set is *infinite* if it is not finite.

Definition 0.9.3 (Countable and Uncountable Sets). A set A is *countably infinite* if $A \sim \mathbb{N}$. A set is *countable* if it is finite or countably infinite. A set is *uncountable* if it is infinite but not countable.

Definition 0.9.4 (Comparing Cardinalities). Write $|A| \leq |B|$ if there exists an injection $f : A \hookrightarrow B$. Write $|A| < |B|$ if $|A| \leq |B|$ but $A \not\sim B$.

Theorem 0.9.5 ($\mathbb{Q}$ is countable). *The set $\mathbb{Q}$ of rational numbers is countable. Go to proof.*

Theorem 0.9.6 (Countable union of countable sets is countable). *Let $\{A_n\}_{n \in \mathbb{N}}$ be a countable family of countable sets. Then $\bigcup_{n=1}^{\infty} A_n$ is countable. Go to proof.*

Theorem 0.9.7 ($\mathbb{R}$ is uncountable). *The set $\mathbb{R}$ of real numbers is not countable. Go to proof.*

Theorem 0.9.8 (Cantor's Theorem). *For any set A , there is no surjection $A \rightarrow \mathcal{P}(A)$. In particular, $|A| < |\mathcal{P}(A)|$. Go to proof.*

Definition 0.9.9 (Function Space B^A). For sets A and B , the *function space* B^A denotes the set of all functions from A to B :

$$B^A := \{f : A \rightarrow B\}.$$

Theorem 0.9.10 (Schröder–Bernstein Theorem). *If $|A| \leq |B|$ and $|B| \leq |A|$, then $A \sim B$.*

Equivalently: if there exist injections $f : A \hookrightarrow B$ and $g : B \hookrightarrow A$, then there exists a bijection $h : A \rightarrow B$. Go to proof.

Proofs